



Compositional changes in thermophilous oak forests in Poland over time: do they correspond to European trends?

Kamila Reczyńska  & Krzysztof Świerkosz 

Keywords

Environmental changes; Forest management; Game browsing; Long-term changes; Semi-permanent plots; Soil Reaction; Species composition; Vegetation

Abbreviations

EIV = Ellenberg Indicator Value; PEV = Potential Evapotranspiration.

Nomenclature

Mirek et al. (2002) for plants; Roleček (2013) for plant communities

Received 23 February 2016

Accepted 11 November 2016

Co-ordinating Editor: Radim Hédľ

Reczyńska, K. (corresponding author, kamila.reczynska@gmail.com),
Świerkosz, K. (krzysztof.swierkosz@life.pl)

Museum of Natural History, Wrocław University, Sienkiewicza 21, PL 50-355, Wrocław, Poland

Abstract

Questions: Have compositional changes occurred in thermophilous oak forests in Poland during the past two decades? What kinds of drivers might be responsible for these changes? Are the thermophilous oak woods still rich in species or have they rather lost their diversity?

Location: Sudetes Mts., SW Poland.

Methods: The vegetation of thermophilous oak forests within four distinct areas was surveyed on 92 semi-permanent plots in the time periods from 1992 to 1994 and from 2010 to 2014. The overall pattern in vegetation changes, trends in species composition and site parameters, as well as the diversity of studied communities were analysed using ordination and permutation techniques.

Results: The species composition of thermophilous oak forests has changed significantly. We recorded a decrease in the share of oligotrophic light-demanding species, whereas mesotrophic light-demanding and nitrophilous species showed an upward trend. Regarding the fourth group of shade-tolerant species, their share varied depending on the site. The analysis of habitat parameters revealed that the studied oak forest communities generally shifted from communities of moderately oligotrophic and mesic character towards forests growing on nutrient-richer but drier habitats. Interestingly, the observed change in species composition was not accompanied by the loss of biodiversity that has generally been reported in other types of thermophilous forests in Europe.

Conclusions: Changes observed within thermophilous and previously coppiced oak forests of southwest Poland are going in a direction which has not been recognized so far in analogous communities. The transition of studied communities is probably a result of the cumulative influence of several drivers, the most crucial of which seem to be soil regeneration, after the period of intensive acidity caused by sulphur emission between 1960 and 2000, and decreasing moisture. The observed changes are independent of the type of forest management (or the lack thereof) within the study sites.

Introduction

Many studies conducted recently in Europe and North America, at different temporal and spatial scales, have confirmed both the relevance and rate of changes observed in forest ecosystems of the temperate zone (Lindner et al. 2010; Bertrand et al. 2011; Verheyen et al. 2012; Dirnböck et al. 2014). Nevertheless, the causes of changes are still not sufficiently understood. This is mainly impeded by an

abundance of drivers that could potentially affect forest communities, as well as the local conditions, geographic position and the history of the study sites. Among the highly important factors, a few are mentioned constantly, e.g. changes in forest management, involving cessation of the traditional methods of management (e.g. Baeten et al. 2009; Hédľ et al. 2010; Verheyen et al. 2012; Kopecký et al. 2013; Müllerová et al. 2015); excessive N deposition (e.g. Økland et al. 2004; Bernhardt-Römermann et al.

2007, 2010; Dirnböck et al. 2014); acidification of soils caused by an increasing concentration of SO₂ (e.g. De Schrijver et al. 2006; Van Calster et al. 2007; Baeten et al. 2009; Hédél et al. 2011); the impact of game (e.g. Kirby 2001; Petřík et al. 2009; Hédél et al. 2010; Boulanger et al. 2014; Frerker et al. 2014) and climate change (Van der Veken et al. 2004; Bertrand et al. 2011; De Frenne et al. 2013). Moreover, interrelations between different drivers are often emphasized as being important, e.g. the simultaneous increase in the amount of N and the acceleration of soil acidification (Thimonier et al. 1994; Diekmann & Dupré 1997; Brunet et al. 1998; Falkengren-Grerup et al. 1998), or the buffering role of the canopy layer, which delays the onset of changes due to both global warming (Bertrand et al. 2011; De Frenne et al. 2013) and excessive N deposition (Verheyen et al. 2012). Such varied results in the research conducted over recent years indicate that the process of change taking place in the forest communities of the temperate zone is very complex, multifactorial and far from having an unambiguous explanation (Vellend et al. 2013).

Against this background, thermophilous oak forests in Central Europe, which have been formed under the influence of a specific combination of abiotic factors as well as traditional management (coppicing, pasturing), have become a particularly interesting research object. These forests, depending on their location, are dominated by *Quercus petraea*, *Q. pubescens* or *Q. robur* and occupy very steep, south-facing slopes with shallow, skeletal soils or plateaus (Ellenberg 2009; Roleček 2013; Reczyńska 2015). They had been used for centuries as coppice forests (Hédél et al. 2010; Szymura 2010), which contributed to their maintaining high species diversity. In the Polish part of the Sudetes (SW Poland), thermophilous oak forests, which occupy approximately 3000 ha (Szymura 2010), have not been coppiced since the end of War World II. At the same time, they are still considered one of the forest communities richest in species in Poland, and their role in the protection of biodiversity is considered to be crucial (Kiedrzyński & Jakubowska-Gabara 2014).

Currently, successional processes that have followed the cessation of traditional management practices are causing the transition of thermophilous oak woods into more closed-canopy forests, resulting in excessive shading of the herb layer, homogenization of species composition and a drastic decrease in species diversity in most forest stands that have been investigated in this regard so far (e.g. Van Calster et al. 2007; Baeten et al. 2009; Hédél et al. 2010; Kopecký et al. 2013). Therefore, we can hypothesize that in the Sudetes Mts., thermophilous oak forests are influenced by the same processes that are observed within similar forests in other parts of the continent. Consequently, oak woods in the Sudetes Mts. should have gradually

shifted from species-rich, heliophilous and oligotrophic communities towards species-poorer, closed-canopy forests growing in moister and nutrient-rich conditions (Hédél et al. 2010; Kopecký et al. 2013).

In the present study we focused on three interconnected issues: (1) have compositional changes occurred in thermophilous oak forests in Poland during the past two decades; (2) what kinds of drivers may be responsible for these changes; and (3) are the thermophilous oak woods still rich in species or have they rather lost their diversity?

Methods

Study area

Our research was conducted in the Sudetes Mts. and their foreland (SW Poland) within four distinct forest complexes (Fig. 1). The study area is characterized by varied geological structure, consisting of rocks of different origin (volcanic, metamorphic, sedimentary), and diverse relief forms such as steep slopes, deep river valleys or resistant to weathering monadnocks. The most common soil types are cambisols (on plateaus) or rankers (on steep slopes). The average annual temperature and rainfall, for period 1980–2010, reach 9.4 °C and 515 mm, respectively (according to CRU TS 2.3 data; <https://crudata.uea.ac.uk/cru/data/hrg/>, accessed 20 January 2016). All the investigated stands are of coppice origin, but they differ among one another in terms of the type of management. Wąwóz Lipa is a strictly protected nature reserve; Grochowa Mt. was partially clear-cut 30 yr ago and since then standard management measures (restocking and thinning) have been introduced; for the most part of the Bazaltowa Mt. and Dobromierz occasional selective logging has been implemented. All study sites are also under a strong influence of large populations of wild browsing ungulates (see Table S1.1 in Appendix S1 for detailed description).

Data collection

We resurveyed in total 92 semi-permanent plots of thermophilous oak forests. The first surveys were conducted between 1992 and 1994 (hereafter 1994) by Kwiatkowski (1995, 1996), Szczęśniak (1997) and Berdowski & Radziuk (1998) and then the relevés were resurveyed by the authors in 2010–2014 (hereafter 2014). During both periods of surveying, relevés were collected according to the Braun-Blanquet approach (Mueller-Dombois & Ellenberg 2002) in patches recognized by the authors of the original relevés to be representative of oak forest communities. Plots were relocated according to the available information about their original position (type of community, slope, aspect). In all cases, a resurvey was conducted within plots adapted to those originally specified in the source data

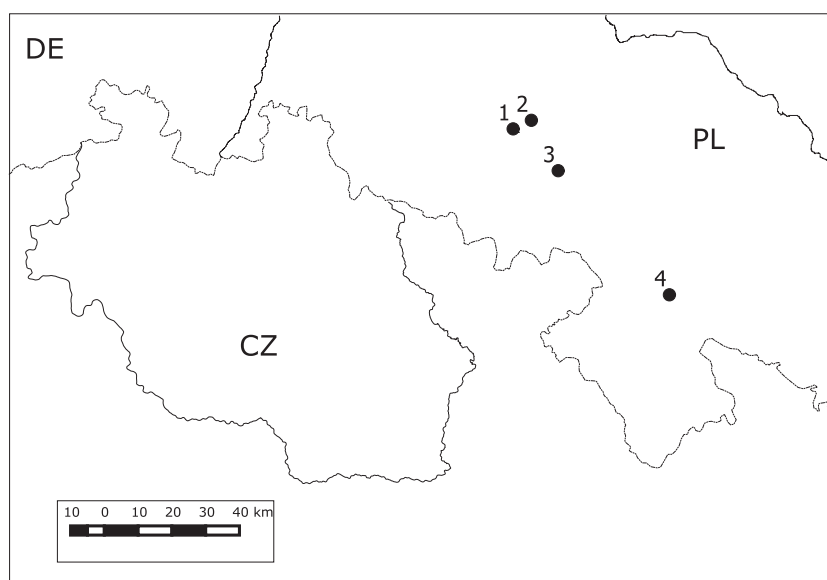


Fig. 1. Distribution of the studied thermophilous oak forest areas in SW Poland (Sudetes Mts. and their foreland): 1 – Wąwóz Lipa, 2 – Bazaltowa Mt. 3 – Dobromierz, 4 – Grochowa Mt.

(100 m², 200 m², rarely 400 m²). In the case of individual plots with unusual area we standardized them to the nearest accepted sizes (80–100 m², 150–200 m², 350–400 m²), hence the average area of the relevé increased between 1994 and 2014 by 52 m². We recorded three layers of the stand (t1 > 20 m, t2 15–19 m, t3 8–14 m), shrub layer (1–7 m), herb layer and moss layer (lichens were excluded). The resurvey was conducted between May and Aug in accordance with time limits in 1992–1994.

Data analyses

In order to prepare the data for further analyses, records of species collected within the relevés for each study site, in different layers, were merged using the JUICE-implemented procedure (Vegetation Science Group, Masaryk University Brno, Brno, CZ) under the assumption that particular covers of species can overlap and that they do so independently of each other (Fischer 2015). Moreover, some species were merged into aggregates (marked as 'agg.'). This concerns *Galium schultesii* agg. (*G. schultesii*, *G. sylvaticum*), *Taraxacum officinale* agg. (= *Taraxacum* sect. *Ruderalia*), *Rubus hirtus* agg. (*R. hirtus*, *R. pedemontanus*, *R. glandulosus*, *R. bellardii*, *R. serpens*) and *Rubus fruticosus* agg. (all *Rubus* sp. except for *R. hirtus* agg., *R. idaeus*, *R. saxatilis*, *R. caesius*).

There were no direct measurements of light, soil and other environmental conditions for all the 92 plots. We therefore calculated mean Ellenberg indicator values (EIVs), weighted by species cover for light (L), temperature (T), moisture (M), soil reaction (R) and nutrients (N). We

used the Ellenberg indicator table (Ellenberg et al. 1991), which was modified by Ch. Berg (unpubl.), implemented in JUICE software.

All analyses were carried out separately for each study site. No square root or logarithmic transformation of cover–abundance scale was used.

Overall pattern in vegetation changes

To compare the vegetation patterns between 1994 and 2014, we performed a principal coordinates analysis (PCoA) using PC-ORD software (MjM Software, Gleneden Beach, OR, US) with the Sørensen (Bray–Curtis) distance measure and species scores calculated using weighted averages. To explain possible differences in species composition between survey periods, EIVs were used. Correlation analysis using sample scores on the first four PCoA axes was performed to identify the statistical relationship between the mean EIVs and the distribution of samples in ordination space. In order to identify the statistical significance of the relationship between EIVs and ordination axes, a modified permutation test with 499 permutations was conducted using the MoPeT_v1.2.r script prepared in R software (R Foundation for Statistical Computing, Vienna, AT; Zelený & Schaffers 2012). The correlation between EIVs and sample scores was expressed by the Spearman ρ coefficient (Table S1.2 in Appendix S1). EIVs were then passively projected into PCoA diagrams (Fig. 2). The significance of temporal changes in EIVs in all study sites was tested using the Wilcoxon signed-rank test. The statistical significance of the overall vegetation change

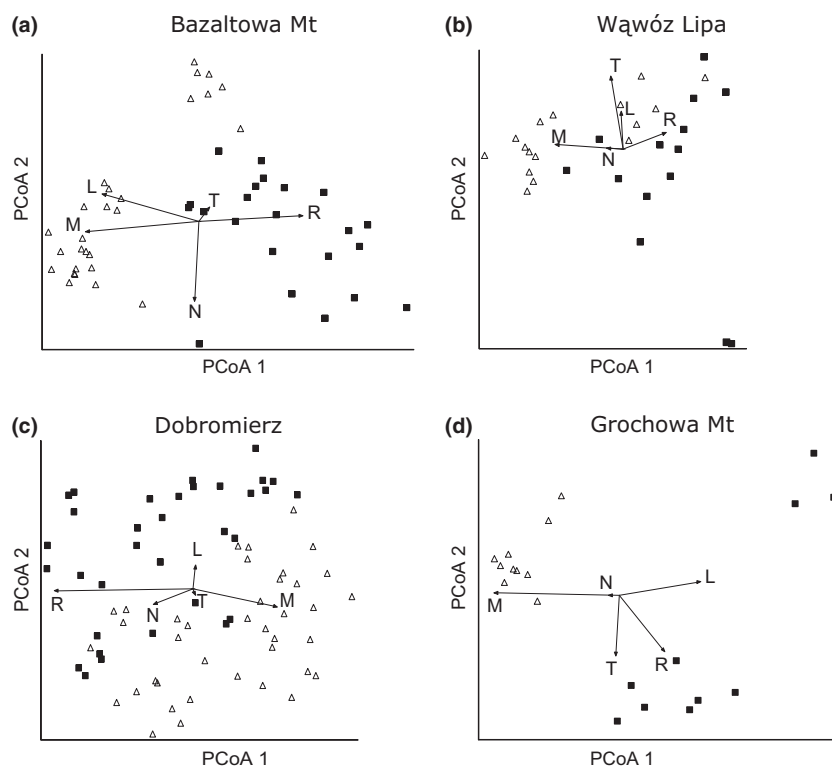


Fig. 2. The shift in species composition during the last 20 yr within four study sites shown in PCoA diagrams. Symbols represent 92 semi-permanent plots for two subsequent periods of time (triangles – 1994, squares – 2014). Vectors of mean EIVs were passively projected into diagrams (L – light, T – temperature, M – moisture, R – soil reaction, N – nutrients). The direction of the vectors indicates the direction of the largest variance of EIVs, whereas their lengths are proportional to their importance.

was tested by PERMANOVA (Anderson 2001). Time was coded as a categorical grouping variable. The significance of changes and variance components, estimated for a random effects model, was assessed by a permutation test with 499 permutations.

Trends of species composition

Trends of studied communities were analysed with an exclusive focus on ecological groups of species defined with EIVs. Within the survey periods for each study site, we distinguished four distinct groups: shade-tolerant species (EIVs: $L \leq 4$, $N \leq 6$), nitrophilous species (EIVs: $L \geq 5$, $N \geq 7$), oligotrophic light-demanding species (EIVs: $L \geq 5$, $N \leq 4$, $R \leq 4$) and mesotrophic light-demanding species (EIVs: $L \geq 5$, $N \leq 6$, $R \geq 5$; Table S1.3 in Appendix S1). The statistical significance of changes in ecological groups of species between survey years was assessed using the SumF method (Warton & Hudson 2004) embedded in PC-ORD. SumF is a permutation test based on aggregated F -statistics. It enables both calculation of a univariate F -statistic for communities as a whole as well as for individual species (Warton & Hudson 2004) and comparison of the

resulting sum to the distribution of F -statistics, which is based on randomizing data under the null hypothesis. F -value shows the significance of changes in the share of species in subsequent periods but without information about the direction of change. Therefore, in order to determine the trend of changes, we used the G value (relative frequency index). In each of the eight sub-data sets (corresponding to the years of research and study sites), we calculated the sum of incidences of species included in the particular ecological group and divided it by the sum of incidences of all species within the sub-data set. The final result was then multiplied by 100.

Additionally, we were interested in the loss and gain of particular species within ecological groups, as the increase and decrease in their abundance may directly indicate different ecological processes occurring over time. For this purpose, we took into account the value of the pseudo F -statistic (calculated with the permutation test) of the species achieved during the SumF analysis, as well as the changes in species frequency in the particular survey period. To define winner and loser species, we set the following parameters: the species had to occur in at least three study sites; the pseudo F -statistic had to be higher than 3

Table 1. The significance of temporal changes in EIVs tested by the Wilcoxon signed-rank test.

	Light	Temperature	Moisture	Soil Reaction	Nutrients
Bazaltowa Mt.	377***	209 ^{n.s.}	321.5***	316**	193.5 ^{n.s.}
Grochowa Mt.	49*	33 ^{n.s.}	55**	54**	29 ^{n.s.}
Wąwóz Lipa	100 ^{n.s.}	92 ^{n.s.}	84.5*	125**	77 ^{n.s.}
Dobromierz	572*	456 ^{n.s.}	709***	613**	410 ^{n.s.}

n.s., not significant; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$.

within at least one study site; and the change in the frequency of the species between the studied periods of time had to be higher than 0.5% in each study site separately. These limitations were due to the fact that we were interested in general trends in the share of more common species rather than the fluctuations resulting from the random occurrences of sporadic species.

Species diversity

Because of different areas of the relevés between survey years, changes in species diversity were assessed based on the first three Hill numbers: $q = 0$ (species richness), $q = 1$ (Shannon exponential diversity) and $q = 2$ (Simpson inverse diversity) computed for each study site (Jost 2006). According to Jost (2006), the exponential of the Shannon entropy is the only diversity measure that can be consistently decomposed into meaningful independent α and β components. However, we also counted the species richness and Simpson diversity due to both their wide use in studies on changes in vegetation and the possibility of comparing our results with previous research. The analysis was performed using ESTIMATES (Version 9, R.K. Colwell, <http://purl.oclc.org/estimates>, accessed 25 January 2016).

Results

Overall pattern in vegetation changes

The species composition of oak forests changed significantly within all study sites (Fig. 2). The distribution of samples along the main PCoA axes indicated clear differences between survey periods on Bazaltowa Mt. and Grochowa Mt., whereas a slightly less visible distinction was recorded in the Dobromierz and Wąwóz Lipa. The first two PCoA axes explained 38.4%, 36.45%, 39.45% and 53.95% of compositional variability within Bazaltowa Mt., Wąwóz Lipa, Dobromierz and Grochowa Mt., respectively.

Correlation analysis of mean EIVs and the main compositional gradients showed that the arrangement of samples along the first two PCoA axes was significantly related to the EIV for soil reaction within all study sites (Table S1.2 in Appendix S1). This suggests that the most important factor responsible for the observed changes in species composition was increasing soil pH. The second variable that determined the distribution of samples was a declining EIV for

moisture, which showed a significant correlation with the first axis and the second axis on Grochowa Mt. and in the Dobromierz, respectively. In the case of Bazaltowa Mt. and Wąwóz Lipa, the second and third PCoA axes were correlated with increasing content of nutrients (EIV for N), which suggests a meaningful influence of local factors on the observed variance.

Analysis of changes in mean EIVs tested by the Wilcoxon signed-rank test revealed that in all study sites the EIVs for moisture and soil reaction significantly decreased and increased, respectively. Similarly, the EIV for light was statistically significant for the most of sites, but the direction of its change was ambivalent – in Bazaltowa Mt. and Wąwóz Lipa it decreased, whereas in Grochowa Mt. and Dobromierz it showed an upward trend (Fig. 2). Changes between the EIVs for nutrients and temperature were statistically insignificant for all study sites (Table 1).

The overall temporal change tested by PERMANOVA was statistically significant ($P < 0.01$) within all study sites (Table 2).

Trends of species composition

The results of the SumF analysis revealed statistically significant changes in all tested ecological groups of species between the survey years for three study sites (Table 3). Only within Wąwóz Lipa were changes in the share of both oligotrophic light-demanding and nitrophilous species not statistically significant.

The direction of changes showed a clear trend for three groups of species (Table 3). The share of oligotrophic light-demanding species has decreased within all study sites. With respect to nitrophilous and mesotrophic light-demanding species, their share has gradually increased over the studied period of time. When it comes to the fourth group of shade-tolerant species, its share has varied

Table 2. The significance of the overall vegetation change tested by PERMANOVA.

Oak forest site	<i>F</i>	<i>P</i>	% of explained variation
Bazaltowa Mt.	12.62	0.002	30.09
Wąwóz Lipa	4.67	0.002	18.67
Grochowa Mt.	7.87	0.002	40.72
Dobromierz	8.98	0.002	16.98

Table 3. The significance of changes in four main ecological groups of species, within all investigated sites, between 1994 and 2014, tested using the SumF method. The direction of changes in the share of ecological groups of species is shown using G value (relative frequency index).

Species Group	Bazaltowa Mt.			Wąwóz Lipa			Grochowa Mt.			Dobromierz		
	F	P	G change	F	P	G change	F	P	G change	F	P	G change
Shade-Tolerant	187.01	0.002	33	64.51	0.004	30	93.76	0.001	−29	91.35	0.001	−21
Mesotrophic Light-Demanding	182.32	0.002	21	92.22	0.008	26	109.80	0.001	171	247.83	0.001	59
Oligotrophic Light-Demanding	63.92	0.001	−24	21.77	0.084	−42	66.62	0.001	−52	85.76	0.001	−28
Nitrophilous	41.82	0.001	73	20.44	0.062	842	28.22	0.009	27	28.22	0.009	5

depending on the site – within Bazaltowa Mt. and Wąwóz Lipa, it has increased; within Grochowa Mt. and Dobromierz, it has shown the opposite trend.

The direction of the abovementioned changes was less clear when we analysed the shares of single species within previously distinguished groups (Table S1.3 in Appendix S1). Within all study sites, there was a marked increase in the share of many shade-tolerant and mesotrophic light-demanding species, as well as some nitrophilous taxa.

Species diversity

Within all study sites, there was a rise in biodiversity regardless of analysed biodiversity index (Table 4). It seems that the main cause of increasing diversity was a simultaneous rise in the share of species included in different ecological groups. Only some oligotrophic species typical of mesic habitats clearly reduced their abundance (Table S1.3 in Appendix S1).

Discussion

Species and vegetation change

The most evident result of our study is that the species composition of thermophilous oak forests has changed significantly over a period of 20 yr. Based on the current state of knowledge, we expected that studied oak woods would probably shift from species-rich, well-lit and oligotrophic communities towards species-poorer, closed-canopy forests growing in moister and nutrient-rich conditions (Jakubowska-Gabara 1993; Hédli et al. 2010; Kopecký et al. 2013). However, the changes that took place within the investigated areas only partially corresponded to the results of the earlier studies. A very clear rise was recorded within both shrub and herb layers of woody species as well as in herbaceous species, which are considered to be diagnostic for thermophilous oak forests (Table S1.3 in Appendix S1). With respect to oligotrophic light-demanding species, there was a distinct difference between the increasing participation of thermophilous and drought-tolerant species and a strongly declining share of species typical of nutrient-poor but mesic habitats.

Sudetic oak forests are also marked by an increase in the number of shade-tolerant or mesotrophic species (e.g. Hédli et al. 2010; Verheyen et al. 2012; Kopecký et al. 2013), as well as nitrophilous ones. The latter grew rather due to the gradual decline in the share of oligotrophic taxa than by their massive penetration into forest communities. This group was still poor in species and did not act as dominant in analysed plots, confirming the results obtained by Dirnböck et al. (2014).

Driving forces of community change

The most important driver causing the shift in species composition of the studied oak forests seems to be a change in soil reaction, caused by an increase in pH. The role of this factor was suggested by the statistical significance of both the correlation of the EIV for soil reaction with the first two PCoA axes (Table S1.2 in Appendix S1) and analysis of temporal changes in this factor within all investigated sites (Table 1). The importance of soil reaction was also confirmed by a substantial rise in the participation of both shade-tolerant and thermophilous species, which prefer habitats with higher soil pH (EIV for $R > 6$). Data on soil pH obtained in the surveyed sites in 1968 (during a particularly intense period of acidification which was attributed to excessive deposition of SO_2 ; Vestreng et al. 2007), in comparison with the data collected in 2015, show a significant increase in the pH of the soils within three out of four investigated sites (there is a lack of archival data for Grochowa Mt.; Table S1.4 in Appendix S1). In research on changes in vegetation, the increasing soil pH has not been recognized so far as a factor – instead of this, negative shifts being attributed to soil acidification were emphasized (e.g. Van Calster et al. 2007; Baeten et al. 2009; Hédli et al. 2011). However, the rise in soil pH, as a result of reduced SO_2 pollution in the 21st century, was recently recorded in the soils of Austrian forest communities (Jandl et al. 2012) and in the soils of England and Wales (Kirk et al. 2010). Although the acidification associated with SO_2 deposition is still treated as one of the main factors affecting vegetation, it should be noted that the European sulphur emissions declined from 55 Tg in 1980 to 15 Tg in 2004 (Vestreng et al. 2007). Thus, the gradual regeneration of

Table 4. Changes in species diversity between years 1994–2014 within four investigated areas of oak forests in the Sudetes Mts. and their foreland.

	Bazaltowa Mt.			Wąwóz Lipa			Grochowa Mt.			Dobromierz		
	1994	2014	Change [%]	1994	2014	Change [%]	1994	2014	Change [%]	1994	2014	Change [%]
Species Richness ($q = 0$)	25.3 ± 7.3	44.9 ± 9.4	77.5	28.8 ± 8.6	39.8 ± 16.5	38.2	24.9 ± 5.6	39.1 ± 5.2	57.0	21.8 ± 7.1	32.4 ± 12.2	48.6
Shannon Exponential	14.1 ± 1.45	33.8 ± 2.4	139.6	16.7 ± 1.76	27.0 ± 2.66	62.0	17.3 ± 1.9	36.7 ± 2.79	112.2	16.9 ± 1.15	33.3 ± 2.46	97.7
Mean ($q = 1$)												
Simpson Inverse Mean ($q = 2$)	4.6 ± 0.21	9.2 ± 0.97	100.9	4.6 ± 0.35	7.4 ± 0.87	62.4	7.4 ± 0.84	15.1 ± 1.84	103.8	5.4 ± 0.34	9.7 ± 1.15	80.4

soils throughout Europe has become possible, even though it takes place much faster on soils rich in carbonates (Jandl et al. 2012). Soils with low buffering properties are vulnerable to the effects of excessive SO₂ deposition for a longer time (Oulehle et al. 2006; Hédél et al. 2011). Interestingly, soils derived from bedrocks with a high content of cations (basalt, green schist, spilite), which the studied thermophilous oak forests develop on, also seem to regenerate relatively quickly.

Forest management is the next factor whose influence may be considered significant for changes observed in the studied oak forests. Our research suggests that the increasing density of the canopy within strictly protected sites supports shade-tolerant species and causes the withdrawal of oligotrophic light-demanding species, which is consistent with the results of studies on other forest communities (Baeten et al. 2009; Hédél et al. 2010; Verheyen et al. 2012; De Frenne et al. 2013; Kopecký et al. 2013). Additionally, the increasing shading may facilitate the accumulation of N (Hédél et al. 2010; Verheyen et al. 2012), thus promoting nitrophilous species and causing an increase in species diversity, as was suggested by Bernhardt-Römermann et al. (2010). On the other hand, similarly to Vild et al. (2013), we also observed an improvement of light conditions and a decline in the share of shade-tolerant species within sites where extensive forest management combined with the removal of single trees was implemented.

Another factor that probably influences the current condition of thermophilous oak forests in the Sudetes is the presence of large populations of browsing wild ungulates – especially roe deer (see Table S1.1 in Appendix S1), which has a number of ramifications for an ecosystem. They are the most important factor leading to reduced recruitment of all tree species (e.g. Petřík et al. 2009), which decreases the density of shrubs and understorey, and prevents the shading of the forest floor. However, although activity of grazers generally promotes light-demanding species it may also intensify eutrophication and negatively influence oligotrophic species (Petřík et al. 2009), as was observed in the present study. The pressure of wild ungulates can also cause a reduction in competition from expansive species with clonal growth, highly sought out by deer (e.g. *Rubus fruticosus*, *Vaccinium myrtillus*, *Hedera helix*, *Urtica dioica*), which enables the development of species sensitive to competition and also may increase species diversity (e.g. Kirby 2001; Frerker et al. 2014). Another consequence of a high abundance of ungulates is the rise in the share of grasses (Kirby 2001), which was also confirmed during our research; among the winners (Table S1.3 in Appendix S1), a substantial part were grasses such as *Brachypodium pinnatum*, *B. sylvaticum*, *Festuca ovina*, *Poa nemoralis*, *Bromus beneckenii*, *Arrhenatherum elatius* and *Millium effusum*. Apart from

grazers, an additional factor that might influence the growth in the coverage of the abovementioned species was increasing N deposition (De Schrijver et al. 2011).

The last factor that might influence studied communities is climate change – one of the most important causes of the transformation of forest ecosystems in the 21st century (e.g. Lenoir et al. 2008; Lindner et al. 2010; De Frenne et al. 2013). One of the measures which may reflect climate change at the local scale is potential evapotranspiration (PEV) – a function of available solar radiation, temperature of the air and evaporating surfaces, air humidity and wind speed (Aussenac 2000). It increases with higher temperatures and winds, and even when precipitation is stable, it exerts a substantial impact on vegetation.

In the Sudetes, between 1980 and 2010, the average annual PEV has increased by $11 \text{ mm}\cdot\text{mo}^{-1}$, whereas the average for Jul has increased by $60 \text{ mm}\cdot\text{mo}^{-1}$ (according to CRU TS 2.3 data; <https://crudata.uea.ac.uk/cru/data/hrg/>, accessed 20 January 2016). Therefore, it is possible that changes in microclimatic conditions could be responsible for the increasing share of xerophilous species resistant to periodic summer droughts, such as *Viscaria vulgaris*, *Festuca ovina*, *Sedum maximum*, as well as some of thermophilous species typical of forests, such as *Digitalis grandiflora*. The results of our PCoA showed that the majority of plots documented in the 1990s had more hygrophilous species than the recent survey. The same was suggested in the analysis of temporal changes in EIVs using the Wilcoxon signed-rank test, which revealed a significant decrease in moisture values within all study sites. However, the role of this factor in transformation of studied communities requires further research.

Trends in diversity of thermophilous oak forests

The most often recorded changes within oak forest communities are the loss of biodiversity through the decrease in the share of light-demanding species, as well as an increase in the number of nitrophilous and shade-tolerant species (Baeten et al. 2009; Hédél et al. 2010; Kopecký et al. 2013). However, in this study, we observed an increase in species diversity regardless of the level q at which this factor was analysed.

Of course, it is possible that the reason for this surprising rise is errors resulting from the imprecise collection of relevés in the past. However, relevés obtained from literature sources were made by four different authors, which reduces the risk of these kinds of mistakes. Differences in the number of species do not result from different areas of the relevés either. The area of relevés documented in literature ranged from 100 to 400 m^2 (mean 178 m^2), whereas replots

were mostly collected on a surface of 200 m^2 – somewhat larger (mean 230 m^2). The average difference in the number of species between plots of thermophilous oak forests should not exceed seven species (average $31.7 \text{ species}/180 \text{ m}^2$ vs $39 \text{ species}/230 \text{ m}^2$), so not more than 23% (Świerkosz 2009). However, the observed increase (between 38.2% and 77.5%, average 55%) in the number of species is twice as high as theoretical values. Moreover, changes in biodiversity seem to be generally resistant to the area of the vegetation plot (see Vellend et al. 2013), as well as to the imprecise location of vegetation plots in repeated studies (see Kopecký & Macek 2015). Of course, there is also a possibility that the species richness of studied communities had gradually decreased till 1992, reaching an absolute minimum, and since then it has progressively increased. This could explain incompatibility of our results with the outcomes of other studies on coppice forests in which the first surveys were conducted in the early 1950s and then were repeated at the beginning of the 21st century (e.g. Baeten et al. 2009; Hédél et al. 2010; Müllerová et al. 2015). Under this assumption, the species richness within study sites, although currently much higher than in 1992–1994, would not achieve its original level when studied communities were used as coppice forests, or even in the post-war period shortly after cessation of the economic use and grazing. Unfortunately, there are no data or basic materials that would enable the analysis of such a scenario.

Conclusions

In previous studies, homogenization and impoverishment of the herb layer due to both the abandonment of traditional management and the conversion of coppice forests towards high forests were recorded. Moreover, in the presented case of the Sudetic oak forests, we are dealing with an opposite trend. Although these communities have not been traditionally managed for a long time, we observed a significant rise in biodiversity, as reflected in an increasing share of thermophilous and/or light-demanding species typical of the *Quercetalia pubescenti-petraeae* or thermophilous grasslands, as well as shade-tolerant species. Thus, the studied oak forests changed significantly from communities of moderately oligotrophic and mesic character towards more heterogeneous forests growing on nutrient-rich but drier habitats. Moreover, the abovementioned transition was recorded within all studied oak forest complexes, regardless of whether they were strictly protected or more heavily managed.

Processes observed within the thermophilous oak woods can be linked to the cumulative influence of several

drivers, most importantly the regeneration of soils, after intensive acidification in the 1980s, and decreasing moisture. The other analysed drivers such as forest management, game browsing and N deposition may have additional impacts on oak forest vegetation also at the local scale.

The example of oak woods in the Sudetes reveals a need for further research on the changes in vegetation of temperate forests that are not schematic and strongly dependent on local conditions.

Acknowledgements

We would like to thank Martin Macek, Radim Hédli, Jörg Ewald and two anonymous reviewers for valuable comments on the manuscript. We thank Kamil Błoński and Henryk Litwin (Regional Directorate of the National Forests in Wrocław), as well as the Bureau of Forest Management and Geodesy in Brzeg for providing archival and contemporary data on soil properties. We also thank Jacek Orda and Kamil Witek for sharing information on the abundance of grazers. Special thanks to Amanda Anthony, who improved our English.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Appendix S1. Supplementary tables to the paper.